

FRAUD DETECTION USING MACHINE LEARNING

ABSTRACT

This project focuses on detecting fraudulent financial transactions using Machine Learning techniques. With the increasing use of digital payment systems, fraud detection has become an important requirement for banking and financial institutions.

The system analyzes transaction patterns and predicts whether a transaction is genuine or fraudulent. Machine Learning models are trained on historical data to identify suspicious behavior automatically.

1. INTRODUCTION

In modern financial systems, millions of transactions occur every day. Among these, some transactions are fraudulent and can cause significant financial losses.

Traditional rule-based systems are not efficient for detecting complex fraud patterns. Therefore, Machine Learning is used to build intelligent systems that learn from past data and detect fraud more accurately.

This project aims to develop a fraud detection system that:

- Analyzes transaction behavior*
 - Identifies suspicious patterns*
 - Classifies transactions as fraud or non-fraud*
-

2. DATASET DESCRIPTION

The dataset contains financial transaction records with multiple features related to sender, receiver, and transaction details.

Key attributes include:

- Transaction type (TRANSFER, CASH_OUT, etc.)*
- Transaction amount*
- Initial and final balances of sender*
- Initial and final balances of receiver*
- Fraud label indicating whether the transaction is fraudulent or not*

The dataset is highly imbalanced, meaning that fraud cases are very rare compared to normal transactions.

3. DATA EXPLORATION (EDA)

Exploratory Data Analysis was performed to understand the structure and behavior of the dataset.

Observations:

- Most transactions are normal, while fraud cases are very few.*

- *TRANSFER and CASH_OUT transaction types show higher fraud probability.*
- *Transaction amount distribution is highly skewed.*
- *Balance changes in accounts play an important role in detecting fraud.*

Visualizations Used:

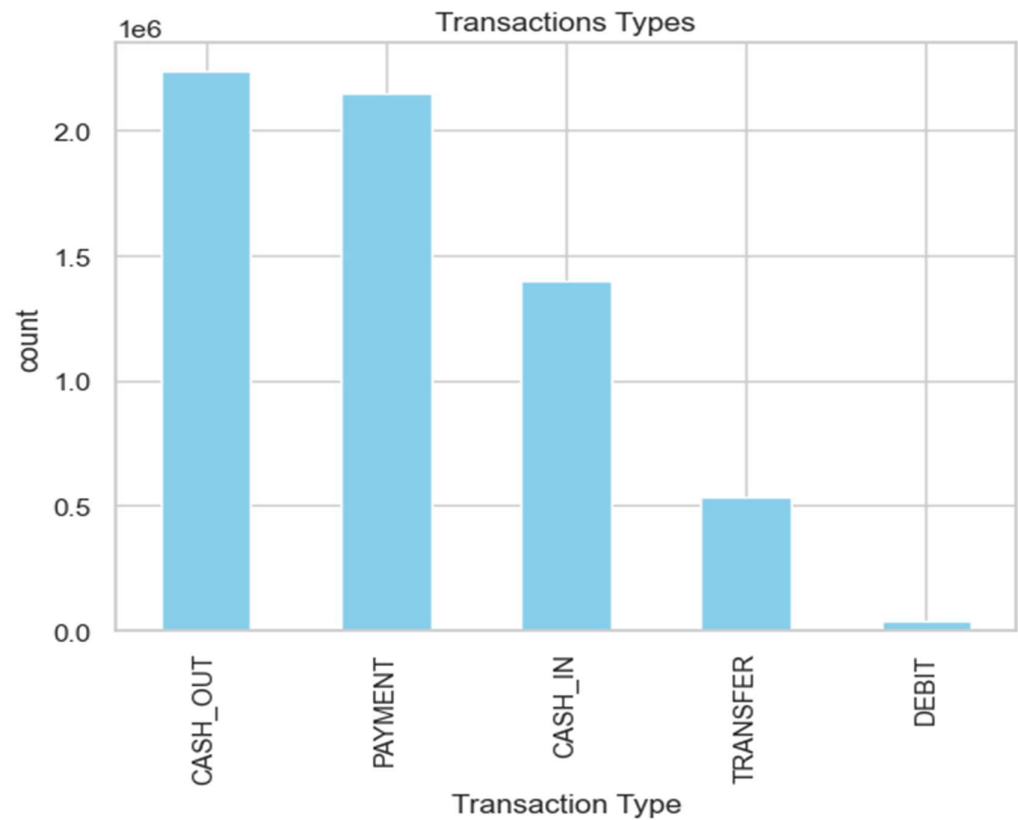
- Transaction type distribution graph
- Fraud rate by transaction type
- Transaction amount distribution (log scale)
- Boxplot comparing fraud and non-fraud amounts
- Correlation heatmap between features
- Fraud occurrences over time

These visualizations helped in understanding patterns and identifying important features.

3. EXPLORATORY DATA ANALYSIS (FULL VISUALIZATION SECTION)

Exploratory Data Analysis (EDA) was performed to understand patterns in the dataset and identify important features for fraud detection.

3.1 Transaction Type Distribution

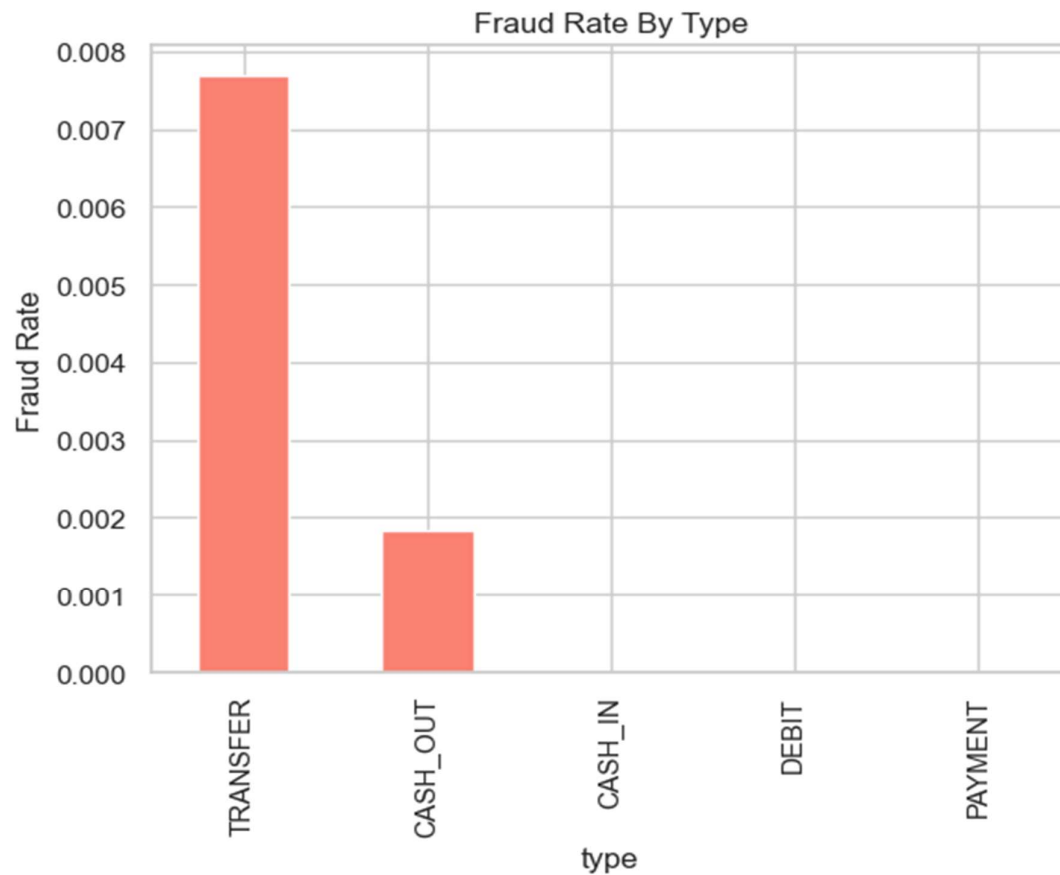


3.2 Fraud Rate by Transaction Type

This graph shows the probability of fraud occurring in each transaction type.

Insight:

- *TRANSFER and CASH_OUT have higher fraud rates*
- *Other transaction types show very low fraud probability*



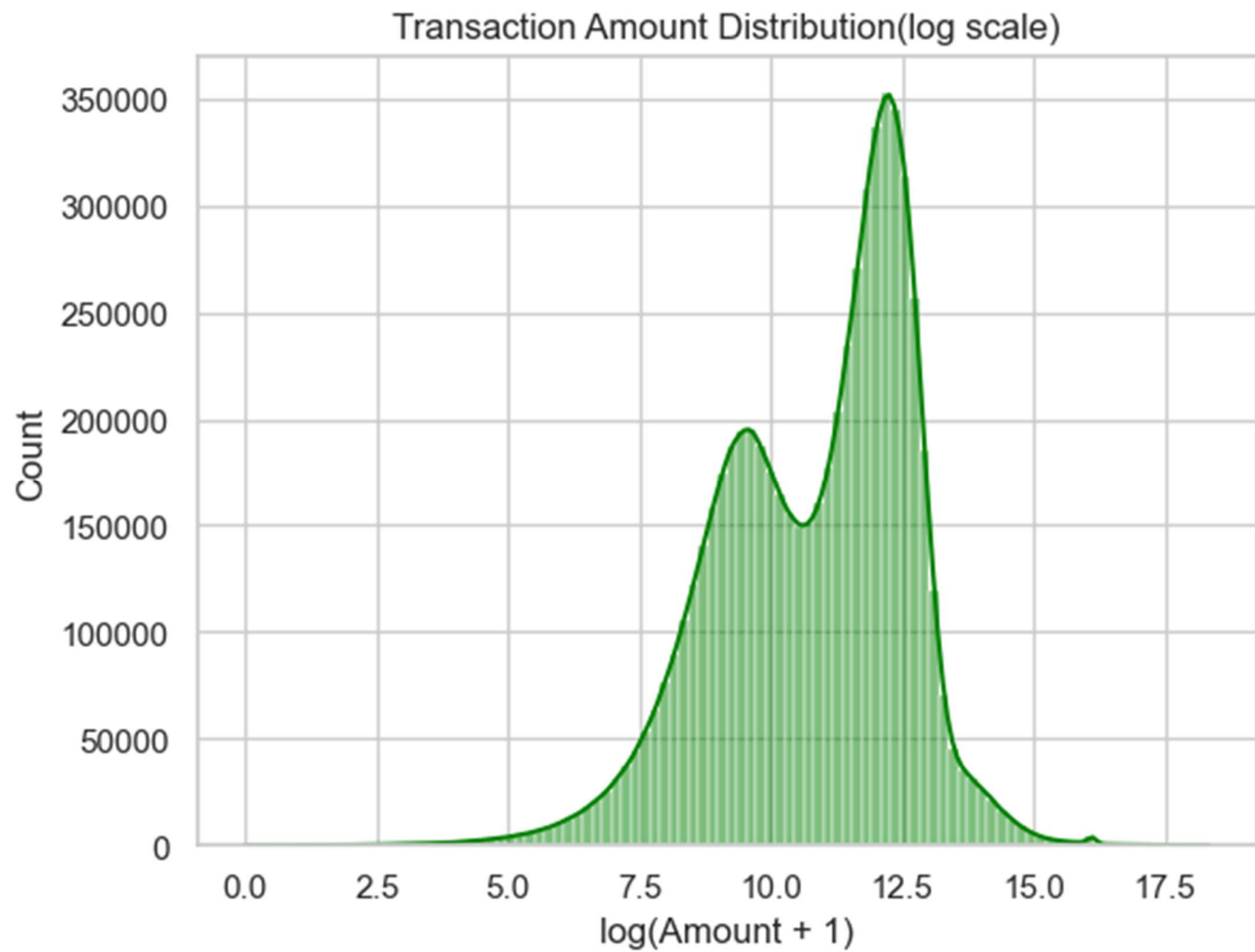
3.3 Transaction Amount Distribution

This visualization shows how transaction amounts are distributed in the dataset using a log scale.



Insight:

- *Most transactions are of small value*
- *Large transactions are rare but more suspicious*

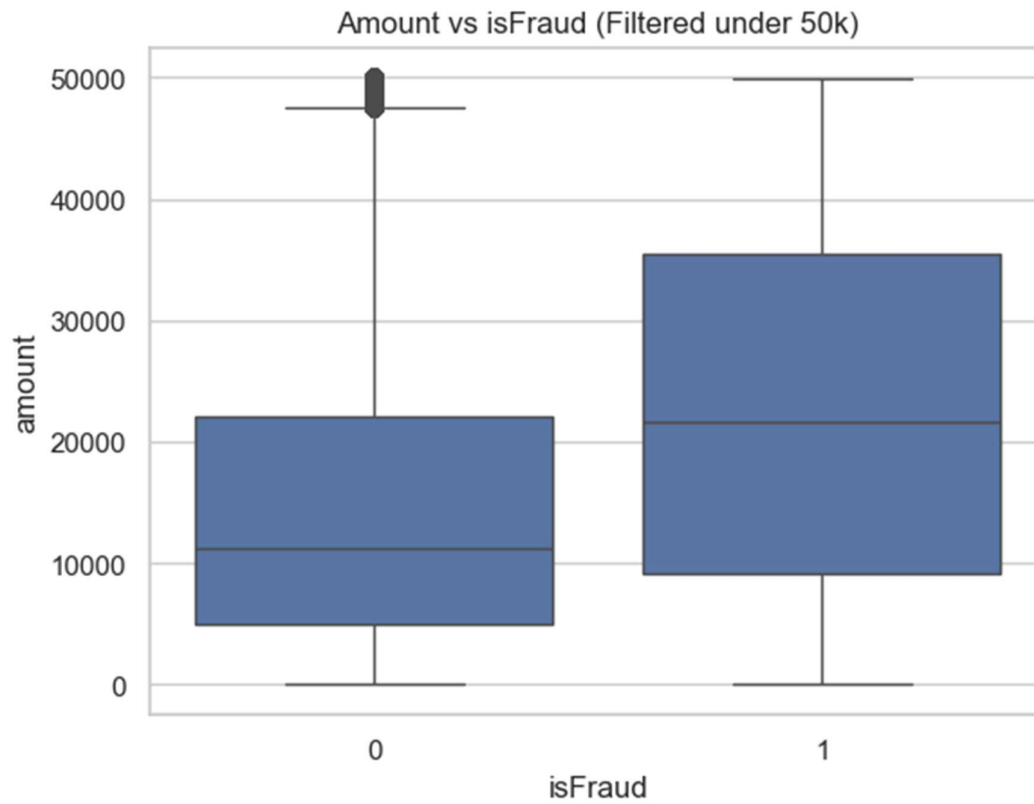


3.4 Boxplot: Amount vs Fraud

This plot compares transaction amounts between fraud and non-fraud cases.

Insight:

- *Fraud transactions sometimes involve unusual amounts*
- *Outliers are present in fraud cases*

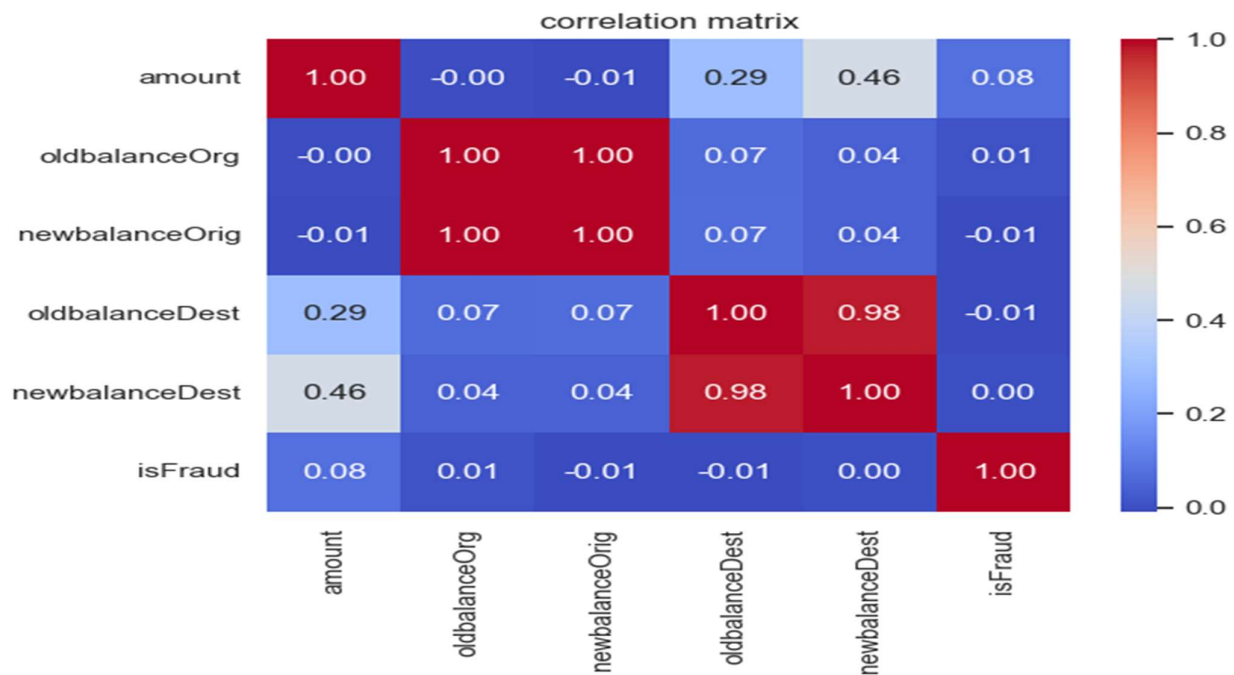


3.5 Correlation matrix

This visualization shows relationships between numerical features.

Insight:

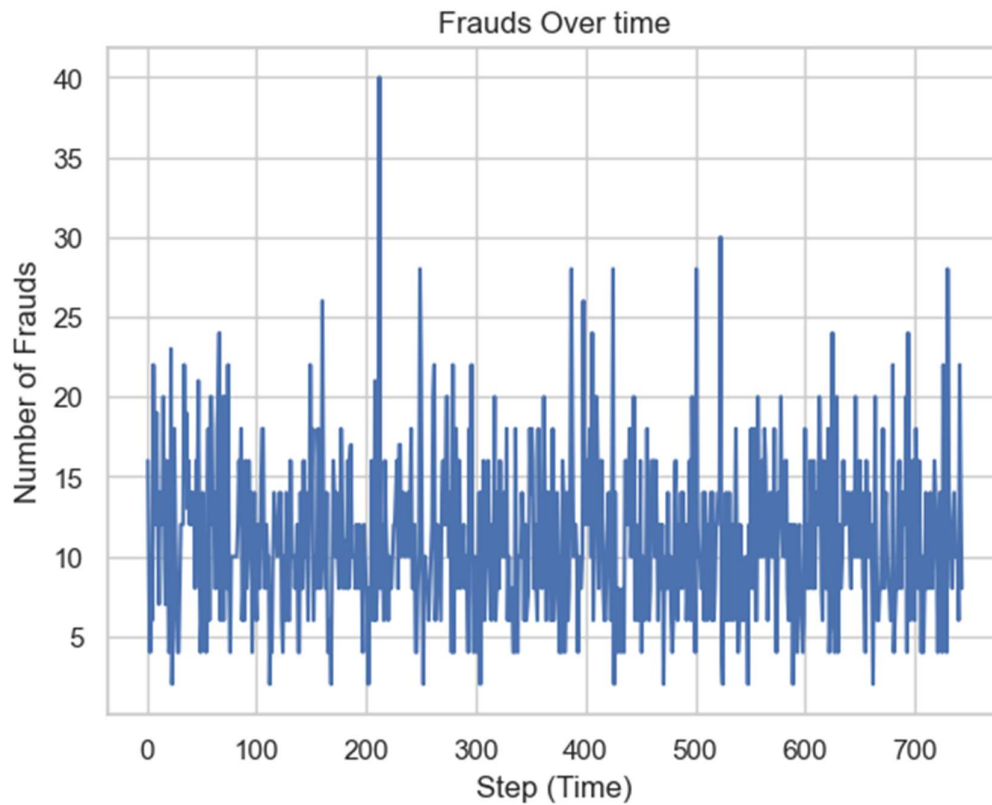
- Strong correlation exists between balance-related features
- Balance differences are important indicators of fraud



3.6 Fraud Over Time (Step-wise Analysis)

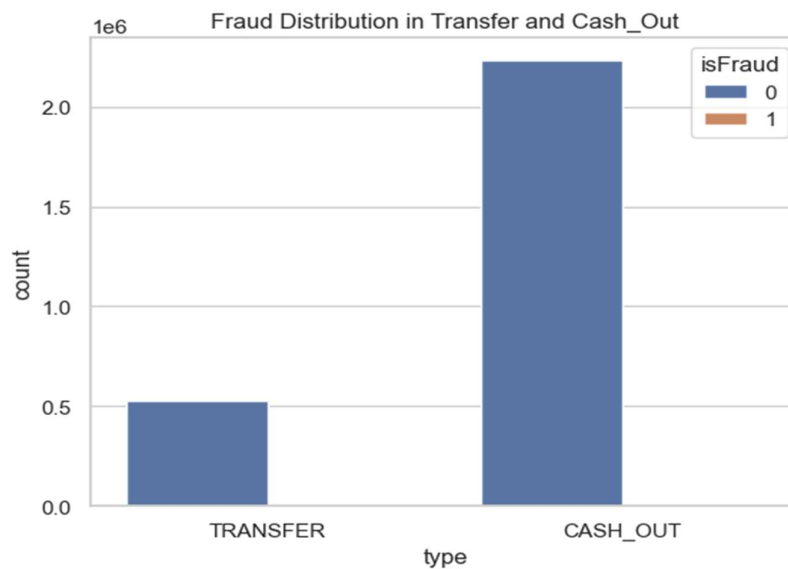
This graph shows how fraud cases are distributed over time steps.

- Fraud is not evenly distributed over time
- Certain time periods show higher fraud activity



3.7 Fraud Transactions in TRANSFER & CASH_OUT

This visualization focuses only on high-risk transaction types.



Insight:

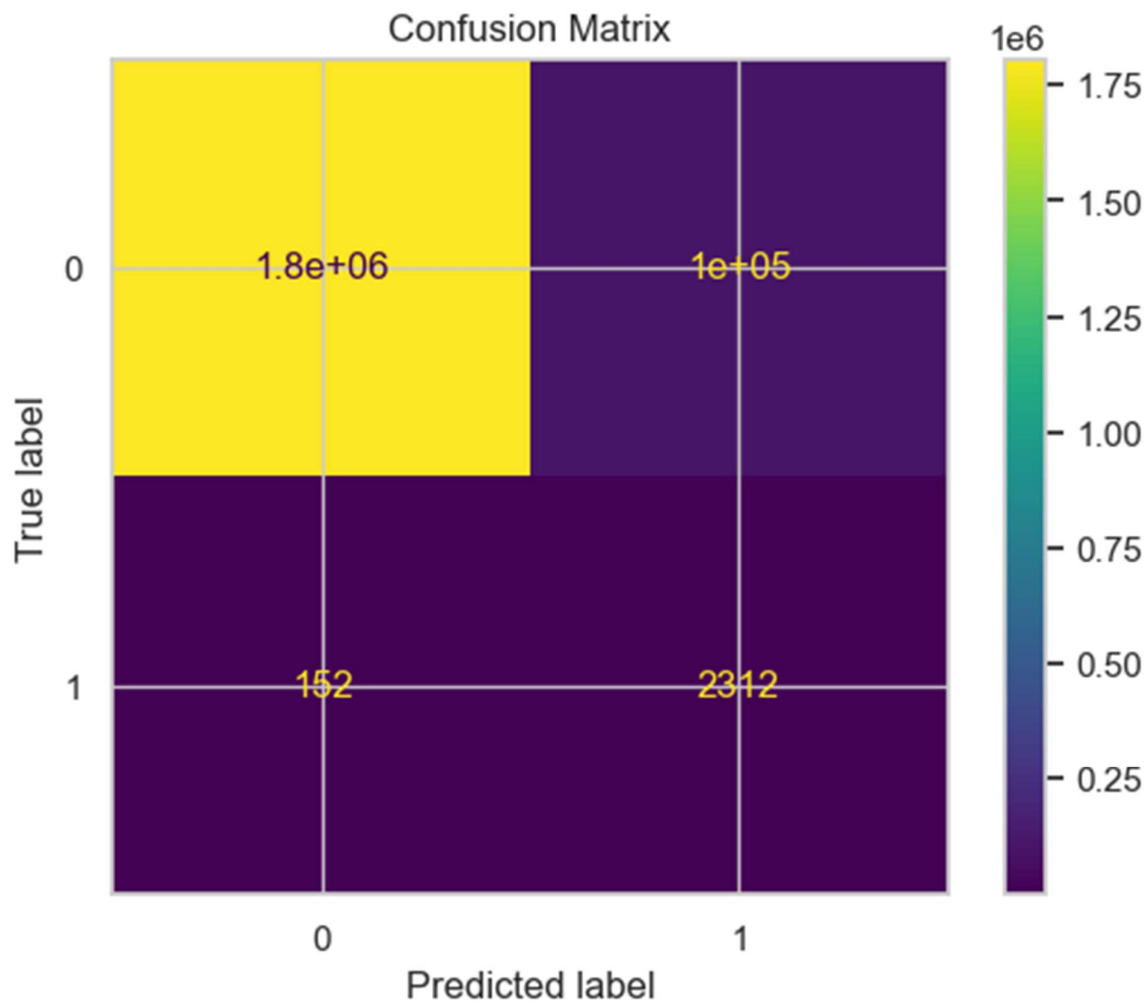
- These two categories dominate fraud cases
- Helps in identifying high-risk transaction groups

3.8 Confusion matrix

This visualization shows model performance by comparing actual and predicted values.

Insight:

- True Positive and True Negative represent correct predictions
- False Positive and False Negative represent errors
- High TP and TN indicate good model performance
- False Negatives are important in fraud detection as they show missed fraud cases

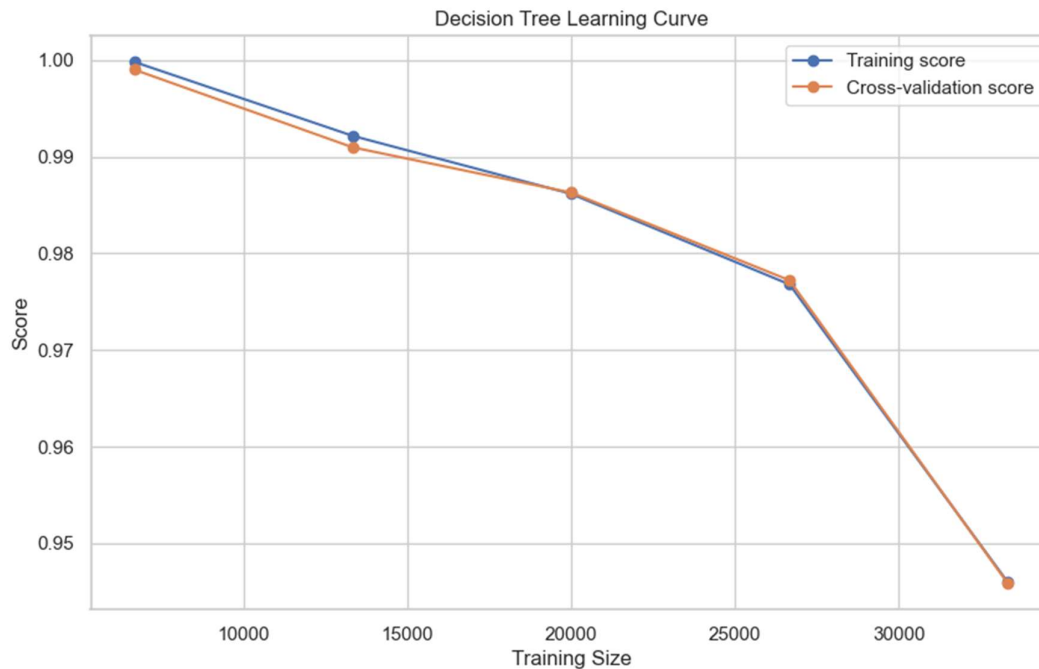


3.9 Decision Tree Learning Curve

This visualization shows model performance with respect to training data size.

Insight:

- Helps detect overfitting and underfitting
- Training and validation scores are compared
- Small gap between curves shows good model performance
- Large gap indicates overfitting



4. FEATURE ENGINEERING

New features were created to improve model performance.

Key idea:

Instead of only using raw values, additional features were derived from balance changes in accounts.

Examples of insights:

- Sudden decrease in sender balance may indicate fraud
- Unusual increase or mismatch in receiver balance is suspicious

These engineered features helped the model understand hidden fraud patterns more effectively.

5. DATA PREPROCESSING

Before training the model, data was cleaned and prepared.

Steps performed:

- Irrelevant columns like transaction IDs were removed
- Numerical features were standardized to bring them to a similar scale

- *Categorical features were converted into numerical format*
- *Data was split into training and testing sets*

This ensured that the model receives clean and structured input data.

6. MODEL BUILDING

Two machine learning models were used:

1. Logistic Regression Model

- *Used as the primary classification model*
- *Works well for binary classification problems*
- *Handles imbalance using class weighting*

2. Decision Tree Model

- *Used for comparison*
 - *Captures complex decision rules*
 - *Helps understand feature importance*
-

7. TRAINING PROCESS

The dataset was divided into training and testing sets.

- *Training data was used to teach the model patterns*
- *Testing data was used to evaluate performance on unseen data*

The model learned relationships between transaction features and fraud labels.

8. MODEL EVALUATION

The performance of the model was evaluated using multiple metrics:

1. Classification Report

It includes:

- *Precision → How many predicted fraud cases were actually fraud*
- *Recall → How many actual fraud cases were detected*
- *F1-score → Balance between precision and recall*

2. Confusion Matrix

A matrix showing:

- *Correct predictions*
- *Incorrect predictions*
- *False alarms and missed frauds*

3. Accuracy Score

Shows overall correctness of the model predictions.

9. MODEL COMPARISON

Two models were compared:

Logistic Regression:

- *Fast and efficient*
- *Good for linear relationships*
- *Performs well with balanced weights*

Decision Tree:

- *More interpretable*
 - *Captures complex patterns*
 - *Can overfit if not controlled*
-

10. KEY FINDINGS

- *Fraud transactions are rare but follow specific patterns*
 - *Certain transaction types are more risky*
 - *Balance changes are strong indicators of fraud*
 - *Machine Learning improves fraud detection accuracy significantly*
-

11. CONCLUSION

The project successfully demonstrates how Machine Learning can be used for fraud detection in financial systems.

The model is able to:

- *Identify suspicious transactions*
- *Handle imbalanced data*
- *Learn from historical transaction behavior*

This system can help financial institutions reduce fraud and improve security.

14. FUTURE SCOPE

The system can be improved further by:

- *Using advanced models like Random Forest or XGBoost*
- *Implementing deep learning techniques*
- *Deploying as a real-time fraud detection system*
- *Integrating with banking APIs for live monitoring*